

Adaptive Civic Routing Intelligence System

Deterministic PostGIS Spatial Containment with Assistive AI for Municipal Governance

Automating civic complaint jurisdiction boundaries, inter-departmental dispatch, and version provenance across Mysuru.

TEAM SPARK | ID: HM26-E9C5

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VERIFIED LIVE MVP & REPOSITORY

Production URL:

<https://adaptive-civic-routing.vercel.app>

GitHub Repository:

<https://github.com/kkaribasappakl-eng/adaptive-civic-routing>

Stack: Node/Express + PostGIS + React/Vite + Socket.IO + Gemini AI

The Problem: Municipal Jurisdiction Mismatch in Mysuru

Core Civic Routing Bottlenecks

1. Complex & Overlapping Municipal Boundaries

Mysuru City Corporation \(\text{MCC} - 65 \text{ wards}\) and MUDA \(\text{urban development authority}\) share irregular, winding border corridors. Citizens cannot tell who owns which street.

2. The Pin-Code & Geocoding Fallacy

Traditional portals rely on 6-digit postal pin codes or commercial reverse-geocoders. Addresses show 'Sayyaji Rao Rd, Mysuru' but lack municipal jurisdiction boundaries.

3. Administrative Ticket Bouncing & Lost SLA

Misrouted complaints take 7 to 14 days to be rejected with 'Not our jurisdiction'. Citizens lose trust; field workers waste hours traveling to incorrect sites.

4. Lack of Historical Provenance During Redistricting

When ward borders delimit, past records are overwritten, destroying audit trails.

Realistic Mysuru Field Scenario

Scenario: Sayyaji Rao Road / Vijayanagar Outskirts Overflow

A citizen spots hazardous dumping near a border drainage ditch at Lat 12.2958 N, Lng 76.6394 E. Under existing portals, the ticket is sent to MUDA Roads. After 9 days, MUDA closes it: 'Under MCC Ward 42 Solid Waste Management'.

Result: Hazard remains unattended for weeks. Zero inter-agency routing.

Who Is Affected Today?

??? 1.3 Million Citizens in Greater Mysuru:

Frustrated by lack of tracking, bureaucratic ping-pong, and delayed fixes.

??? Civic Officers & Dispatch Operators:

Overwhelmed by manual triage phone calls and ambiguous boundary maps.

??? Municipal Engineers & Sanitation Crews:

Assigned tickets outside their physical jurisdiction; wasted fuel and field hours.

Target Metric: Reduce initial jurisdiction dispatch from 72 hours to 0 seconds.

Target Users: Tailored Workflows for Municipal Governance

CITIZEN PERSONA

Citizen

Device: Mobile Web Browser / Smartphone

OPERATIONAL CONTEXT

Everyday residents walking or driving in Mysuru encountering potholes, overflows, and street lighting issues.

WHY EXISTING WORKFLOWS FAIL

Existing municipal portals demand complex ward numbers and department dropdowns, creating a high barrier to entry for citizens.

OUR SYSTEM ADVANTAGE

Single GPS tap on interactive Leaflet map. Assistive Gemini AI classifies categories and suggests the correct department.

OPERATOR PERSONA

Civic / Operator Officer

Device: Municipal Desktop / Command Tablet

OPERATIONAL CONTEXT

City Corporation triage officers and control room supervisors managing city-wide incidents.

WHY EXISTING WORKFLOWS FAIL

Drowning in misdirected complaints; spends 60% of work hours re-routing tickets to the correct department.

OUR SYSTEM ADVANTAGE

Dedicated Review Workspace with automated PostGIS routing. Ambiguous border areas are highlighted for review.

FIELD CREW PERSONA

Field Worker / Supervisor

Device: Rugged Mobile / Department Dashboard

OPERATIONAL CONTEXT

Junior engineers, health inspectors, and sanitation supervisors executing physical repairs.

WHY EXISTING WORKFLOWS FAIL

Dispatched to repair sites only to discover upon arrival that the road belongs to a different ward.

OUR SYSTEM ADVANTAGE

Guaranteed jurisdiction containment: tickets delivered to departmental queues are routed to the correct field crew.

Our Solution: Automated 5-Step End-to-End Civic Journey

The 5-Step Deterministic Routing Pipeline

- 1

Citizen Intake & Proximity Check

Leaflet + ST_DWithin

Citizen drops GPS pin on Leaflet map. Proximity query checks for duplicate complaints within 100m in past 7 days without block
- 2

Assistive AI Category Suggestion

Gemini 1.5 Flash

Google Gemini API (gemini-1.5-flash) analyzes description. Suggests controlled category; transparent manual fallback if uncont
- 3

Deterministic PostGIS Containment

PostGIS ST_Covers

PostGIS runs ST_Covers against active version polygons. Mathematically resolves Authority (MCC vs MUDA). Zero AI guessin
- 4

Department Dispatch & SLA Clock

category_mappings

Relational mapping table pairs (authority_id, category) to assign responsible department. Category-specific SLA target timestan
- 5

Human Review & Immutable Ledger

routing_decisions FK

Out-of-boundary coordinates quarantine into Operator Review queue. All decisions locked permanently to active jurisdiction ver

VERIFIED APPLICATION EVIDENCE



Real Intake Evidence: Sayyaji Rao Road Pothole Hazard

??? Filename: complaint-1789889476281-75d0c6cb42b882d0.png

??? Geotagged coordinates: Lat 12.2958 N, Lng 76.6394 E

??? AI Suggested: POTHOLE (Confidence: 0.94) | Source: AI_SUGGESTED

??? PostGIS Resolution: MCC Central Zone 1 -> MCC Roads Dept (ROUTED)

Architecture: Complete End-to-End System Separation

STRICT SEPARATION LAW: AI assists citizen issue classification. PostGIS and relational rules strictly decide legal jurisdiction & department.

CLIENT & PRESENTATION

??? React 18 SPA + Vite 6
??? Leaflet 1.9 + React-Leaflet
??? OpenStreetMap SRID 4326 tiles
??? Tailwind CSS 3 Dark Palette
??? Lucide React Icons
??? Citizen Complaint Form
??? Operator Review Workspace
??? Boundary Version Manager
??? Analytics Spatial Dashboard
??? Audit Log & Provenance Viewer
??? Live Socket.IO Toast Gateway

API GATEWAY & AUTH

??? Node.js v22 + Express v4
??? JWT Authentication Service
??? Role-Based Access Control:
- CITIZEN (Public Intake)
- OPERATOR (Triage/Claim)
- ADMIN (Boundary Delimit)
??? Multer Upload Engine (5MB cap)
??? Sanitized file hashing
??? MIME check (JPEG, PNG, WebP)
??? Gemini API Integration:
- gemini-1.5-flash endpoint
- Non-faked manual fallback

SPATIAL ENGINE & POSTGIS

??? PostgreSQL 18 + PostGIS 3.6
??? SRID 4326 WGS 84 Coordinates
??? MultiPolygon Ward Geometries
??? Point-in-Polygon: ST_Covers
??? Boundary Edges: ST_Touches
??? GiST Spatial Indexing:
- idx_complaints_location
- idx_jurisdictions_boundary
??? Single Active Version Index:
- uq_single_active_version
??? Immutable Decision Ledger:
- routing_decisions (FK locked)
??? category_department_mappings

REAL-TIME & AUDIT LOG

??? Socket.IO v4 WebSocket Hub
??? Partitioned Room Channels:
- privileged_operators
- role:CITIZEN (live updates)
??? Events: complaint:created,
complaint:status_changed
??? SLA Deadline Calculator
??? Escalation Event Service
??? Immutable Audit Trigger:
- audit_logs table
- Postgres trigger forbids
UPDATE / DELETE
??? 100% Tamper-Proof Trail

Hard Constraints: Honest Evaluation vs. Production Realities

CIVIC CONSTRAINT	STATUS	OUR ACTUAL IMPLEMENTED APPROACH	HONEST TECHNICAL LIMITATION
1. Take Reports	HANDLED	Mobile web intake with Leaflet GPS pin-drop. Real PostgreSQL persistence with unique HM-CIV-2026-XXXXX codes. 5MB photo/MTE check. Proximity Duplicate check (100m/7d).	Resolution: Single active version constraint (the single active version) be quarantined
2. Jurisdiction	HANDLED	PostGIS SRID 4326 MultiPolygons for MCC and MUDA wards. Deterministic ST_Covers point-in-polygon	Resolution: Single active version constraint (the single active version) be quarantined
3. Priority & SLA	HANDLED	Automated SLA target deadlines calculated upon routing based on category severity (complaint_sla_rules)	SLA State as log SUBMITTED > ROUTED > IN PROGRESS > RESOLVED
4. Bad Input	HANDLED	Bounding box coordinate validation (Mysuru Lat 12.2-12.4, Lng 76.5-76.8). Non-image files rejected by MIME. Unmapped coordinates quarantined to HUMAN REVIEW	Mites. Unmapped coordinates quarantined to HUMAN REVIEW; relies on citizen
5. Offline	PARTIAL	System degrades gracefully: if Gemini AI is down, manual category unlocked; if DB down, 503 status returned, frontend displays clear offline/disconnected banners	no client when internet drop

Key Technical Decisions: Rationale & Engineering Trade-Offs

DECISION Q1

Deterministic PostGIS Geometry Containment vs. Probabilistic LLM Text Routing

Why Chosen Over Alternatives:

Municipal authorities (MCC vs. MUDA) possess statutory legal boundaries. Text heuristics and LLMs hallucinate on border routing.

Engineering Trade-Off & Mitigation:

Requires verified GIS shapefiles and spatial indexing. Coordinates outside mapped polygons cannot be auto-routed and are quarantined into manual review.

DECISION Q2

Immutable Jurisdiction Versioning vs. Mutable In-Place Boundary Overwrites

Why Chosen Over Alternatives:

Mysuru ward boundaries periodically delimit (e.g. MCC expanding into MUDA peripheral layouts). New tickets route against MUDA.

Engineering Trade-Off & Mitigation:

Relational schema complexity and storage overhead of retaining retired geometries. Mitigated by atomic version transition transactions and partitioning.

DECISION Q3

In-Database Synchronous Spatial Routing & Municipal Scale Architecture

Why Chosen Over Alternatives:

In-database PostGIS execution ensures immediate ACID consistency at ticket creation (POST /api/complaints), eliminates external joins.

Engineering Trade-Off & Mitigation:

Concurrent spatial queries during civic storm surges can burden database CPU. Roadmap: (1) Bounding-box pre-filtering (&&); (2) Range table.

Proof It Works: Genuine Application Evidence & Verification

1. GENUINE APPLICATION COMPLAINT EVIDENCE



Verified Citizen Evidence Upload in Working Application

??? Proves: Working photo evidence upload, MIME validation, & storage

??? Source: server/uploads/complaints/complaint-1789889476281-...png

??? Location: Lat 12.2958, Lng 76.6394 | Sayyaji Rao Road Corridor

??? Status: Verified in live database, post-processed by Gemini AI & PostGIS

2. Deterministic Routing Receipt & Department Dispatch

Route: GET /api/complaints/:id/routing | Live Citizen Tracking View

What This Proves:

Shows HM-CIV-2026-XXXXXX ticket bound to MCC Zone 1 -> MCC SWM via ST_Covers under active version MYS_2026_V2 (ACTIVE)

Live Verification: Open <https://adaptive-civic-routing.vercel.app> and submit ticket on map.

3. Operator Review Workspace & Boundary Disambiguation

Route: /operator or /reviews | Role: OPERATOR / ADMIN

What This Proves:

Displays active triage queue, SLA countdown timers, and manual override controls for out-of-boundary quarantined cases.

Live Verification: Use Demo Login -> Select Operator in deployed application header.

4. Immutable Version Ledger & Audit Provenance

Route: GET /api/audit/logs | /audit Tab | Role: ADMIN

What This Proves:

Tamper-proof PostgreSQL ledger recording ROUTING_EXECUTED events bound to MYS_2026_V2 (ACTIVE) while preserving MYS_2026_V1 (ARCHIVED)

Live Verification: Database triggers strictly prevent UPDATE or DELETE modifications.

Limitations & Scale: Honest Assessment at Municipal Volume

1

MultiPolygon CPU Load

CURRENT LIMITATION

Synchronous point-in-polygon queries (ST_Covers) executed across high-vertex

WHY IT MATTERS AT SCALE

Under severe monsoon storm surges (thousands of simultaneous geotagged civic

ENGINEERED MITIGATION & ROADMAP

1. Bounding-box pre-filtering (&&) to discard 98% of wards in memory before ST_

2

Client Offline Sync

CURRENT LIMITATION

Complaint submission currently requires active cellular/WiFi data connection. No l

WHY IT MATTERS AT SCALE

Citizens in rural border zones (Chamundi Hill base, peripheral layouts) with interm

ENGINEERED MITIGATION & ROADMAP

Implement Service Worker Background Sync API paired with client-side IndexedD

3

Single-Instance DB

CURRENT LIMITATION

A single PostgreSQL instance handles transactional intake, PostGIS spatial computa

WHY IT MATTERS AT SCALE

Heavy analytical spatial queries (rendering 10,000 incident heatmaps) can compete w

ENGINEERED MITIGATION & ROADMAP

Deploy read-replica architecture: direct transactional intake to primary DB, offload Ge

What’s Next & Formal AI Tool Disclosure

3 Concrete Municipal Next Steps

- 1

Ward-Level Municipal Pilot (MCC Zone 1)
Deploy with MCC Zone 1 field workers (Sayyaji Rao Road corridor) for a 30-day live trial to evaluate dispatch latency and field worker acceptance metrics.
- 2

Cadastral GIS Boundary Expansion
Ingest survey-grade shapefiles from Karnataka State Remote Sensing Applications Centre (KSRSAC) for all 65 MCC wards and peripheral MUDA layouts.
- 3

Kannada Vernacular Voice & Text Input
Integrate speech-to-text models to allow Kannada-speaking citizens to lodge verbal complaints, automatically classified into the controlled municipal taxonomy.

Phase 2 Milestone: Automated cross-departmental escalations via WhatsApp Business API.

Formal AI Tools Disclosure & Links

AI Technologies Utilized in Development & Runtime:

- 1. Google Gemini API (gemini-1.5-flash):**
Used strictly as assistive AI to suggest civic categories from description text.
AI NEVER makes jurisdiction, municipal authority, or dispatch decisions.
- 2. Antigravity AI Assistant:**
Used for codebase architecture planning, test automation, and code review.

All civic routing decisions are 100% deterministic PostGIS spatial SQL.

PROJECT ACCESS & REPOSITORIES

Live Working MVP URL:
<https://adaptive-civic-routing.vercel.app>

GitHub Repository:
<https://github.com/kkaribasappakl-eng/adaptive-civic-routing>

Thank You Judges | HackMysuru 1.0 | Team Spark (HM26-E9C5)